Recoil and explosions

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use momentum conservation for an isolated system.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 2.*
15 kg·m/s left.

02. Worked example: explain the model
Initial total zero; after arrows +4 and -4 kg·m/s cancel.

03. Read and interpret the model
Equal magnitudes, opposite directions; total stays zero.

04. Apply the physics
Momentum 4 kg·m/s right needed; speed 2 m/s.

05. Explain your reasoning
Stored internal energy can become kinetic energy; momentum conservation does not require kinetic energy conservation.

06. Review: connect an earlier idea *Review: Unit 2, Day 9.*
Equal fall times; faster ball travels 6 times farther horizontally.

07. Exit check
6 kg·m/s right.

08. STRETCH: test the model
Second mass 3 kg; equal opposite momenta +6 and -6 kg·m/s.

Recoil and explosions

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Use momentum conservation for an isolated system.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 2.*
v=-5 m/s; after reversal p=+15 kg·m/s.

02. Worked example: explain the model
Initial total zero; after arrows +4 and -4 kg·m/s cancel.

03. Read and interpret the model
Equal magnitudes, opposite directions; total stays zero.

04. Apply the physics
60 v +24=0; v=-0.40 m/s.

05. Explain your reasoning
Stored internal energy can become kinetic energy; momentum conservation does not require kinetic energy conservation.

06. Review: connect an earlier idea *Review: Unit 2, Day 9.*
Speed 20 m/s. Need an initial position plus time and motion history/conditions; velocity alone is insufficient.

07. Exit check
v=-6 m/s; KE 81+54=135 J from released internal energy.

08. STRETCH: test the model
Person -0.5 m/s; relative board speed 3.5 m/s. Velocities in conservation must use one frame.